

Chat Room Project

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September 22, 2026

This project, from a high-level point-of-view, is an application that allows users to create chat rooms with each other. They can chat with others in their respective room through a text-based "terminal" display.

The details:

1. Coded in C++ with Qt's libraries/APIs.
2. UI resembles a terminal
3. User account creation/verification
4. Chat room creation/joining/leaving
5. Messages

Attributes include: person sending it, chat room sent to, time stamp, and a string containing message content.
6. Utilizes OO structure

Users, messages, and chat rooms will all have their respective classes, inheriting and being the parent to other specific classes like a Moderator class.
7. Utilizes network programming

Users can join/leave a server. This is how the chat room is made possible (communication among multiple computers). A server will maintain all currently connected users.
8. Utilizes parallelism/concurrency

A thread pool will handle the many requests to the server by executing tasks in parallel through multiple threads of control. For example, if two chat rooms are active at the same time, they should be operating simultaneously, especially since there data and interfaces are conflicting with each other.
9. includes chat history

Will store data in (?) (DB or file or in server)